

ESI Data Center

Architecture Overview

Spine-Leaf fabric, EVPN/VXLAN overlay, and multi-domain workload isolation.

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01 Introduction

This document describes the architecture decisions for the ESI data center redesign and their implications. The platform accommodates distinct operational workloads—pedagogical computing, staff and administrative services, shared storage, high-performance AI/GPU computing, DMZ services, and the seasonal BAC Orientation platform—each requiring independent scaling, failure isolation, and security boundaries.

1.1 Operational Scope

The data center serves five distinct operational areas:

- **Pedagogical Compute:** Supervised lab servers, student VMs, and learning platforms, treated as untrusted endpoints.
- **Production Services:** Identity, DNS/DHCP, monitoring, and HR/Finance systems requiring high availability and strict access controls.
- **Shared Platform Services:** Ceph-backed block, filesystem, and object storage for virtualized and AI/HPC workloads.
- **Performance-Sensitive Compute:** Bare-metal GPU nodes and SLURM scheduling with RDMA east-west and low-latency requirements.
- **Special-Purpose Edge Services:** DMZ workloads and the seasonal BAC Orientation platform, isolated from production routes with dedicated external connectivity.

1.2 Core Architectural Principles

The redesign is guided by four foundational principles:

- **Reasoning clarity:** Every mechanism has an explicit role, bounded scope, and understood failure mode.
- **Additive Growth:** Capacity expands by adding Leaf pairs, server nodes, or VRF instances without disturbing existing workloads.
- **Performance Determinism:** Oversubscription ratios, link speeds, and transport features are assigned by workload profile, not applied uniformly.
- **Structural Isolation First:** Security boundaries are established through network structure—VRFs, absent routes, dedicated pods, and independent OOB management—before firewall policy; authorized cross-domain paths are centralized at the Edge Pod firewall for stateful inspection and enforcement.

All paths from untrusted zones to production services traverse the Edge Pod firewall. No direct topology adjacency exists between different operational domains; structural isolation precedes and supplements firewall policy at every boundary.

02 Network Topology

The production fabric is a two-tier Clos (Spine-Leaf) topology with redundant Layer 3 uplinks from every Leaf to both Spines; there are no Leaf-to-Leaf links, and Layer 2 switching is confined within each pod boundary. The complete topology is shown in Figure 1.

2.1 Fabric Devices

Device Names	Associated with	Role and Function
S01, S02	Spines	Shared transit layer; AS 65000; underlay and EVPN reachability distribution points
L01, L02	Edge Pod	ISP edge, border routers, firewalls, DMZ, and seasonal BAC Orientation
L03, L04	Core Pod	Core infrastructure and administrative service hosting
L05, L06	Storage Pod	Ceph-backed shared storage cluster
L07, L08	Compute Pod	General-purpose virtualized multi-tenant compute
L09, L10	Ultra Compute Pod	Non-blocking 100 Gb/s GPU and HPC pod
OOPS	Out-of-Band Switch	Dedicated management switch, independent from EVPN fabric

2.2 Fabric Grouping Principles

- **Pod redundancy:** Every Leaf pair defines one pod and provides redundant dual-homing for all attached servers. Failure of one Leaf in a pair does not interrupt server connectivity.
- **Spine minimization:** The two Spines provide 1+1 active-active redundancy for transit reachability: both carry live traffic under normal conditions via hash-based ECMP across Leaf uplinks. Loss of one Spine preserves end-to-end connectivity while halving aggregate uplink bandwidth; BFD-triggered BGP withdrawal removes the failed Spine from ECMP groups.
- **Edge Pod exposure:** The Edge Pod is deliberately the only pod exposed to ISP churn, DDoS absorption, public services, and cross-domain firewalling. This concentrates external risk in a single operational domain.
- **Out-of-band independence:** The OOPS management switch is physically separate from the EVPN fabric and remains the guaranteed recovery path when production control-plane systems become unavailable or compromised.

Current production scale is 2 Spines, 5 Leaf pairs (10 production Leaves), and 1 OOB switch; the OOB network is the guaranteed recovery path when production routing is unavailable.

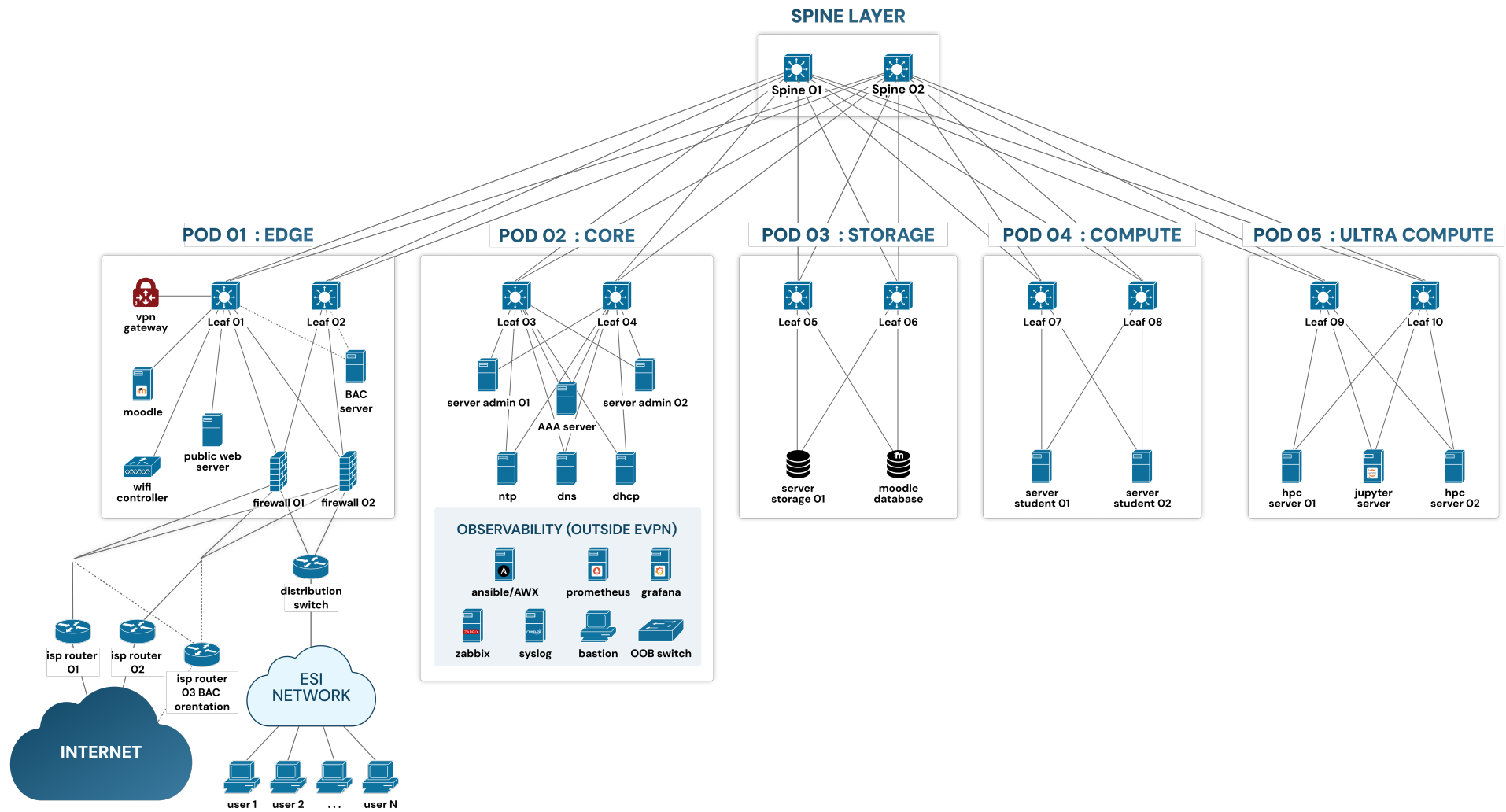


Figure 1. Spine-Leaf production topology, including the Edge, Core, Storage, Compute, Ultra Compute, and Out-of-Band management domains.

03 Structural Constructs

The architecture treats network behavior, operational role, and security context as three *independent* concerns. Pod design addresses hardware constraints and traffic profiles; cluster design addresses orchestration and workload management; communication domains address isolation and controlled inter-system communication. This independence allows each dimension to scale without constraining the others or requiring fabric redesign.

3.1 Core Constructs

Construct	Definition
Pod	A physical attachment zone: one Leaf pair, homogeneous hardware constraints, and a single traffic class
Cluster	A logical operational unit: shared orchestrator, common execution environment, independent of network topology
Communication Domain	A routing boundary: one VRF instance, one L3VNI, and structural security isolation via routing table structure

Clusters span pods transparently to the orchestrator when growth demands it. Security boundary changes require only a VLAN-to-VRF reassignment at the serving Leaf; physical topology and cluster type remain unchanged.

3.2 Pod Architecture

Five production pods serve distinct operational roles:

Pod	Sub. Ratio	Intent
Edge	2:1	North-south traffic, DMZ policy enforcement, and seasonal BAC Orientation
Core	3:1	Always-on infrastructure and privileged administrative services
Storage	1:1–1.5:1	Ceph-backed storage systems and sustained high-throughput data flows
Compute	4:1	General-purpose virtualized multi-tenant workloads
Ultra Compute	1:1	GPU and HPC execution with 100 Gb/s interconnect and RDMA

3.3 Cluster Types and Orchestration

Four cluster types are deployed across the pods: **SVC** (Standard Virtualization Cluster), **SBC** (Standard Bare-metal Cluster), **ACC** (Accelerated Compute Cluster), and **DSC** (Distributed Storage Cluster).

Type	Orchestrator	Posture
SVC	OpenStack + KVM	OpenStack/KVM multi-tenant virtual machine hosting; supports mixed Staff and Pedagogy workloads.
SBC	None	Bare-metal special-purpose servers
ACC	SLURM + Kubernetes	Bare-metal accelerated compute: SLURM manages HPC batch jobs and GPU reservations; Kubernetes manages containerised AI services on the same nodes.
DSC	Ceph	Dedicated storage cluster

3.4 Pod and Communication-Domain Mapping

The current mapping between pods and their primary cluster and communication-domain assignments is shown below. This allocation reflects present-day operational requirements; the model itself does not depend on a one-pod-per-role topology.

Pod	Primary Cluster	Current Communication Domains
Edge	SBC (+ public SVC)	Orientation, Public
Core	SVC	Staff, Administration
Storage	DSC	Staff
Compute	SVC	Staff, Pedagogy
Ultra Compute	ACC	Staff, Pedagogy

Every server is triple-classified by *pod* for traffic behavior and failure domain, by *cluster* for orchestration and operational procedures, and by *communication domain* for routing and security policy; domain membership is assigned by VLAN-to-VRF mapping at the attached Leaf, not by cluster type.

04 Protocol Stack

The protocol stack is a unified underlay-overlay model built on underlay simplicity for predictability, overlay isolation for tenants and security domains, and fabric-wide consistency for observability.

4.1 Stack Layers

Layer	Technology and Intent
Macro-segmentation	VRF per security domain, one L3VNI per domain to enforce routing isolation
Overlay Control	MP-BGP EVPN for scalable, distributed MAC/IP learning and loop-free reachability
Overlay Data	VXLAN over UDP for hardware-independent encapsulation
Underlay Control	eBGP with BFD for sub-second failure detection and convergence
Physical Underlay	Two-tier Clos with /31 point-to-point links; 9000-byte MTU on all fabric links. Jumbo-frame hosts use 8950 bytes to account for VXLAN overhead; standard hosts remain at 1500 bytes.

4.2 Design Principles

The stack is guided by five key principles:

- **Symmetric IRB with anycast gateways:** Every Leaf acts as a first-hop anycast gateway, distributing inter-subnet forwarding across the fabric without single-point bottlenecks.
- **Distributed MAC/IP learning via EVPN:** Type 2 routes suppress ARP flooding and distribute MAC/IP bindings; Type 5 routes carry inter-subnet and authorized cross-domain reachability.
- **BFD on every BGP session:** BFD targets sub-second failure detection with a 300 ms detection window, enabling fast recovery without waiting for BGP hold-timer expiry.
- **Head-end replication for unknown unicast and multicast:** Ingress Leaf replication eliminates PIM in the

overlay, simplifying operations and reducing control-plane state.

- **Dual-homing with LACP and ESI:** Servers dual-home via LACP bond with ESI to both Leaf pairs, remaining active if one Leaf fails.

Leaf-to-Spine eBGP sessions carry both underlay and EVPN address families, with Spines preserving the originating Leaf next hop so VXLAN tunnels terminate directly on the VTEPs.

Detailed stack inventory including ASN assignments, loopback addressing, EVPN route type usage, and underlay addressing conventions is documented in Appendix A.

05 Macro-Segmentation

Security is enforced at the *routing layer* through communication domains implemented as Virtual Routing and Forwarding (VRF) instances. Each communication domain has a completely independent routing table. By default, no inter-domain routes exist—not because a firewall blocks them, but because the routing paths do not exist in the fabric. A missing route is a stronger guarantee than a firewall rule: there is nothing to misconfigure, no policy to inadvertently omit.

When communication between communication domains is permitted by architecture, traffic must traverse the Edge Pod firewall pair for stateful Layer 7 inspection, policy enforcement, and logging. The routing structure guarantees the path through the firewall; the firewall guarantees only authorized access and complete observability.

5.1 Communication Domains

Five communication domains divide the fabric according to operational trust and threat levels:

Communication Domain	L3VNI	Architectural Intent
Pedagogy	10000	Student workloads, labs, and project environments; treated as untrusted by default
Staff	20000	Shared production platform services, compute control plane, and storage consumers
Administration	30000	High-sensitivity institutional systems with tighter operational and access controls
Orientation	40000	Seasonal BAC platform, deliberately isolated from all production routes
Public	50000	Internet-facing services and DMZ with zero internal RFC1918 route imports

5.2 Architectural Decisions

Four core design decisions enforce structural isolation:

- **Routing as the primary barrier:** VRF isolation ensures structural separation before any firewall policy is evaluated. This defense-in-depth approach prevents misconfigured firewall rules from creating unintended paths.
- **Public communication domain has no internal routes:** The Public communication domain (DMZ) imports no

internal RFC1918 prefixes. A DMZ compromise cannot reach internal networks because no routing path exists, independent of firewall state.

- **Symmetric IRB with centralized control:** Leaves route only the VNIs they locally serve and use the per-domain L3VNI for symmetric IRB across the fabric. Every authorized path between communication domains is hairpinned through the Edge firewall pair.
- **Leaf-assigned communication-domain membership:** Communication-domain membership is enforced by VLAN-to-VRF assignment at the attached Leaf pair according to the workload's security classification, not by cluster type alone.

Detailed VNI/subnet mappings and per-Leaf VRF configurations are in Appendix B. This section focuses on communication-domain boundaries and the principle of structural routing-layer isolation.

06 Services

Services are placed by traffic profile, bandwidth requirement, and failure-domain intent so each pod retains predictable operating characteristics.

6.1 Pod Roles and Service Placement

Each pod serves a distinct operational purpose:

Pod	Sub. Ratio	Role	Service Posture and Rationale
Edge	2:1	Border and policy enforcement	North-south ingress/egress, inter-domain control points, and seasonal BAC edge exposure
Core	3:1	Shared services	Identity, DNS/DHCP, orchestration, observability, and business applications
Storage	1:1 to 1.5:1	Data services	Ceph-backed block, file, and object services for platform consumers
Compute	4:1	General virtualization	Multi-tenant VM hosting for pedagogy and standard production compute
Ultra Compute	1:1	AI/HPC execution	High-throughput bare-metal GPU workloads with loss-sensitive transport controls

6.2 Service Placement Decisions

- **Edge Pod:** Border routers, the HA firewall pair, public web services, the public Jupyter ingress proxy, and the BAC Orientation nodes terminate here so external routing churn, DDoS exposure, and all authorized communication-domain crossing policy remain confined to L01/L02.
- **Core Pod:** The Core Pod hosts Staff-domain infrastructure services such as identity, DNS/DHCP, NTP, orchestration, observability, JupyterHub control services, and internal portals. HR/Finance remains in the Administration domain on the same Leaf pair but isolated by VRF.
- **Storage Pod:** The Storage Pod exposes Ceph RBD, CephFS, and RGW services in the Staff domain for VM volumes, shared notebook/HPC files, image repositories, and backup exports. Bulk storage and backup flows remain adjacent to the storage cluster so they do not disturb Core Pod latency. Detailed backup re-

tention and off-site replication policy belongs in the detailed architecture or appendix.

- **Compute Pod:** OpenStack/KVM hosts mixed Staff and Pedagogy VMs for efficiency, but routing context remains isolated through VLAN-to-VRF binding at the Leaf, maintaining communication-domain separation regardless of hypervisor placement.
- **Ultra Compute Pod:** The Ultra Compute Pod runs bare-metal ACC nodes with SLURM scheduling, 100 Gb/s uplinks, and RDMA-oriented congestion controls scoped to L09/L10. Interactive notebooks run as scheduled jobs on these nodes rather than as permanent control-plane services.
- **BAC Orientation:** Dedicated bare-metal servers remain logically distinct and are powered off and disconnected outside the operational season; when activated they use the Orientation communication domain and the dedicated third ISP fibre only.

Notebook access is split by function: public HTTPS terminates on the Edge proxy, JupyterHub control services remain Staff-internal in Core, notebook kernels execute on Ultra Compute through SLURM, and user files come from Storage. This preserves separation between ingress, control state, execution, and persistent data.

07 Campus Internet Access and Wireless Infrastructure

Campus internet and wireless infrastructure are maintained outside the data center fabric; the DC exposes only tightly controlled service handoffs and must not become the campus switching fabric, AP access layer, or internet-transit path.

7.1 Design Decisions

Element	Decision and Rationale
Campus Internet Path	Campus internet access remains on the campus distribution path, with the Sophos/campus edge filter inline there rather than in the data center fabric.
Campus WiFi APs	Never terminate directly on a data center Leaf. APs remain on dedicated campus network switches separate from production.
Campus Handoffs	Limited to explicitly published DC services only; no AP termination, VLAN extension, or internet breakout.
WiFi Controller	VM hosted in the Core Pod SVC environment in the Staff communication domain, providing centralized management.
WiFi Management Path	L01 only, over VLAN 120/VNI 10120 in a local Wireless Control VRF, isolated from fabric-wide production domains.
Route Scope	Local handoff/link route plus one static /32 to the controller; no default route.
Access Policy	Edge Pod firewall allows only the campus management subnet to reach controller ports; no campus prefix imported into Staff.

08 Permitted Communication

Communication between domains is intentionally sparse: reachability is determined solely by VRF routes, and all authorized cross-domain paths are firewall-anchored. Any boundary not listed in this section is closed.

Source Comm. Domain	Dest. Comm. Domain	Policy	Authorized Scope and Constraints
Pedagogy	Staff	Yes	Restricted to explicitly published Staff-side services, including Moodle in LMS-STAFF and shared DNS/DHCP/NTP endpoints in CORE-INFRA; no blanket Pedagogy-to-Staff reachability.
Administration	Staff	Yes	Restricted to core infrastructure prefixes in CORE-INFRA, such as identity, DNS, and NTP services consumed by Administration workloads.
Staff	Administration	Yes	Restricted to explicitly published Administration-side application prefixes, primarily HR/Finance services in HR-FINANCE.
S/P/A	Public	Yes	Staff, Pedagogy, and Administration reach published Public-domain service VIPs only, such as DMZ web services and user-facing ingress gateways. This is north-south service publication through the Edge firewall.
Public	Staff	Single proxy	Jupyter ingress gateway reverse-proxy traffic only, constrained to the gateway /32, JupyterHub service /32 or VIP, and required HTTPS/WebSocket/API ports.
Public	Other internal	No	Generic DMZ hosts do not import internal routes and have no authorized path to internal communication domains outside the single Jupyter proxy exception.
Pedagogy	Administration	No	No path exists from untrusted student workloads to privileged institutional systems.
Orientation	Other Comm. Domains	No	No route exists to any internal communication domain; when active, the platform's only external path is the dedicated third ISP fiber.
Other Comm. Domains	Orientation	No	No inbound production routes exist toward the Orientation communication domain.
Wireless Control	Staff	One-way	WiFi controller management only; destination is constrained to a single /32 through the local L01 VRF, and no campus prefix is imported into any fabric-wide communication domain.

8.1 Communication Model

All cross-domain traffic hairpins through the Edge Pod firewall for stateful inspection and logging; intra-domain traffic stays within its VRF without firewall traversal; only explicitly published service prefixes are reachable across any boundary.

The only Public-to-Staff exception is the Jupyter reverse proxy, which does not expose the Hub database, SLURM control plane, Storage Pod, or any generic Staff prefix. Wireless Control remains a local L01 VRF with a single controller /32 path and no campus prefix imported into any fabric-wide domain. Permitted and denied paths are fully captured in the table above.

09 Physical Architecture

Six racks are aligned to operational pod domains, creating clean failure domains and deterministic link-speed classes for consistent performance and growth.

9.1 Rack Layout and Purpose

Rack	Role	Architectural Purpose
R01	Network Core/Edge	Spine pair, Edge Leaf switches, border routers, and HA firewalls
R02	Core Pod	Institutional and control-plane services, bastion access, monitoring and security anchors
R03	Ultra Compute Pod	GPU infrastructure with dedicated 100 Gb/s footprint and thermal/power isolation
R04	Storage Pod	Ceph cluster and storage-facing east-west traffic concentration
R05	Compute Pod	General compute servers and shared hypervisor footprint
R06	BAC/OOB	Seasonal BAC bare-metal servers and out-of-band management network termination

9.2 Physical Design Rules

The fabric follows four design rules to maintain consistency and predictability:

- **Fixed speed classes:** 100 Gb/s links serve Spine↔Edge, Spine↔Ultra Compute, and GPU downlinks; 25 Gb/s for standard server-facing and pod uplinks; 10 Gb/s for border-router attachment.
- **Cabling model:** Direct-attach copper (DAC) is the default for in-room connections. External handoffs use media-appropriate links: single-mode fiber for ISP handoffs, copper for local wireless microwave links. This avoids unnecessary transceiver complexity.
- **Out-of-band discipline:** Every switch and server management interface is terminated on the dedicated OOB network (172.16.0.0/24), which is independent from EVPN and VXLAN forwarding, except for a tightly scoped TACACS+ LDAP path to the OpenLDAP OOB interface (TCP/389). This provides recovery access if the production control plane becomes unavailable.
- **Campus and wireless boundary:** Campus internet access and AP traffic remain on the campus distribution layer, while WiFi controller management uses a dedicated bounded path on L01 with no campus prefix imports into the Staff domain.

Detailed port maps and BOMs are outside this overview.

10 Security

Security layers structural routing isolation with Edge, host, management-plane, and observability controls; the threat model is summarized below.

10.1 Threat Model

The architecture defends against twelve distinct threat classes:

ID	Threat	Primary Target
T1	External intrusion	Edge Pod and DMZ
T2	BGP route injection	Routing fabric
T3	BGP table exhaustion	Edge peers and Leaves
T4	VXLAN frame injection	Overlay isolation boundaries
T5	East-west lateral movement	Peers within one communication domain
T6	Cross-domain movement	Communication-domain VRF boundaries
T7	Unauthorized admin access	Switches and servers
T8	BAC data tampering	Orientation platform
T9	Insider or rogue admin	Configuration plane
T10	Undetected intrusion	Monitoring and detection plane
T11	Physical access	Hardware and cabling
T12	Clock desynchronization	Forensics and authentication

10.2 Defense Rings

Five defense rings, plus supporting controls and BAC-specific controls, address these threats:

Control	Scope and Implementation
Perimeter	North-south and authorized cross-domain paths traverse the Edge Pod firewall pair for stateful Layer 7 inspection.
BGP hardening	All external and internal sessions use TCP MD5; ISP peers accept default-route-only import, no RFC1918 export, and max-prefix guard with a threshold of 4 prefixes per peer (a default route plus minimal operational headroom).
Control-plane ACLs	Switch CPU and tunnel endpoints accept BGP TCP/179 and BFD UDP/3784 from fabric loopbacks only, VXLAN UDP/4789 from Leaf VTEPs only, and SSH from Bastion 172.16.0.50 only.
Bastion access	Switch and server administration uses ED25519 keys only, disabled passwords, AllowUsers, MaxAuthTries 3, and LoginGraceTime 30.
Host micro-seg.	Each server role uses nftables with baseline INPUT DROP, ESTABLISHED, RELATED, Bastion-only SSH, and explicitly opened services.
Visibility	Suricata on a dedicated IDS node consumes the Edge mirror feed; syslog is retained 90 days plus a one-year RGW archive; Chrony keeps time skew under 1 s; OOB stays isolated from production routing.
BAC-specific	Orientation systems stay powered off and disconnected 10 months per year, use a zero-route VRF, dedicated ISP fiber, bare metal only, TPM-backed secure boot, and locked power/boot controls.

10.3 Host-Level Access Control

Each host role has a baseline deny-all firewall policy with explicit service exceptions:

Role	Comm. Domain	Services Beyond Baseline
SVC Lab VMs	Pedagogy	None (isolated by comm. domain)
SVC Infra VMs	Staff	DNS/UDP 53 and DHCP UDP 67/68 from internal ranges
NTP Server	Staff	UDP/123 from internal ranges
HR/Finance	Administration	HTTPS from Staff via Edge firewall policy
AI/HPC Research	Staff	TCP/8080 for notebook workflow frontends (SVC VMs hosting research-facing services in the Compute Pod)
ACC Bare Metal	Staff	TCP/6006 for TensorBoard-style visualization (physical ACC GPU nodes in the Ultra Compute Pod; distinct from the SVC research VMs above)
Services Portal	Staff	TCP/80 and 443 from internal clients
DMZ Web Services	Public	TCP/80 and 443 from internet via Edge firewall
SBC Nodes	Orientation	No inbound service exposure when offline
Syslog Collector	Staff	TCP/514 and UDP/514 from infrastructure

10.4 Design Principles and Invariants

Key invariants: Public imports no internal routes; OOB is separate except for the scoped TACACS-to-LDAP path; structural isolation precedes firewall policy; legacy TCP MD5/TACACS protections should move to stronger options when device support permits; secrets and production changes are centrally audited; Phase 1 administration is on-site via Bastion/OOB only.

10.5 Third ISP Physical Connectivity

The BAC Orientation platform uses a separate third ISP fiber terminated in the Orientation VRF, with no route import into production domains. The path is enabled only during the BAC operational window and administratively shut down afterward.

10.6 Security Validation

The architecture is validated against representative structural and failover scenarios:

Validation	Target	Pass Condition
Comm.-domain isolation	VRF boundaries	Unauthorized communication-domain pairs experience 100% packet loss
VXLAN injection	UDP/4789 ACL	Crafted server-sourced VXLAN traffic is dropped silently
BGP authentication	TCP MD5	Session drops immediately with wrong key, recovers when fixed
Max-prefix guard	External peers	Session tears down when BGP table floods
IDS alerting	Edge mirror feed	Scan or exploit signatures produce alerts on the IDS node
Bastion enforcement	SSH access	No SSH service banner from non-Bastion IP addresses
DMZ isolation	Public comm. domain	No route exists from DMZ hosts to internal RFC1918 prefixes
OOB isolation	Management network	No production route reachability from OOB segment
BFD failover	Spine uplink loss	Fabric reconverges in under 1 second (BFD timers: min-rx/min-tx 100 ms, multiplier 3; 300 ms detection window)
ESI failover	One Leaf failure	Server LACP bond stays active, traffic continues
Backup restore canary	SVC + ACC data restore	Quarterly canary restore succeeds; both backup_restore_ok gauges remain 1
Pipeline provenance	AWX production jobs	Every production job has non-empty pipeline-ID metadata
BAC isolation pre/post	Orientation VRF	Pre-activation and post-deactivation probes from Staff to Orientation gateway remain unreachable

10.7 Operational Observability Posture

Observability is treated as a security and availability control: alerts must map to a threat class or availability invariant, with detailed telemetry sources documented elsewhere.

11 Quality of Service

QoS follows the pod model: each pod has a distinct congestion profile, so critical and loss-sensitive traffic is protected before bursty or bulk traffic. The policy keeps fabric behavior deterministic rather than treating all flows equally.

11.1 Objectives

Objectives are to protect control-plane, BAC, AI/HPC, and interactive service traffic while demoting backup, replication, image sync, and unclassified bulk flows.

11.2 DiffServ Class Model

Prio	Class	DSCP	Queue Behavior	Min BW	ESI Examples
1	Network Control	CS6, 48	Strict priority	Not shared	eBGP, EVPN transport, BFD, LACP, switch control traffic
2	Real-Time	EF, 46	Strict priority, rate-limited	Not shared	Reserved for future voice or video
3	Critical Applications	AF41, 34	High weighted queue	20%	BAC Orientation, critical admin transactions
4	Research / HPC	AF31, 26	High weighted queue	25%	AI/HPC, latency-sensitive storage I/O
5	Interactive / Academic	AF21, 18	Medium weighted queue	20%	LMS-STAFF, SERVICES-WEB, CORE-INFRA, STUDENT-TP
6	General Student	AF11, 10	Low weighted queue	10%	Student projects and general browsing
7	Bulk / Backup	CS1, 8	Scavenger queue	5%	Backup, replication, image sync, software updates
8	Best Effort	DF, 0	Best effort	Residual	Unclassified traffic

QoS classification follows service role as well as VNI, so one segment may carry multiple traffic classes.

11.3 Classification and Marking

Leaves mark DSCP at ingress based on interface, VNI membership, and service role; storage flows are marked by function so latency-sensitive I/O and backup replication receive different treatment.

11.4 VXLAN DSCP Preservation

The fabric uses VXLAN, so DSCP must be preserved across encapsulation. VXLAN interfaces are configured with `tos inherit`, copying the inner DSCP value into the outer header. This allows the spine layer to schedule traffic based on the underlay DSCP value without inspecting the inner payload.

11.5 Queueing and Scheduling on the Spine

Spines schedule traffic according to the DSCP values set at Leaf ingress, using the queue behaviors and minimum-bandwidth guarantees defined in the class model.

11.6 Congestion Management: ECN and PFC

ECN, and PFC only where RDMA requires it, are scoped to AI/HPC rather than enabled fabric-wide.

11.7 Enforcement Points

Enforcement Point	Function
Leaf Ingress	Classify by interface, VNI, and service role; set DSCP before fabric transit
Spine Transit	Queue and schedule already-marked traffic across equal-cost paths
Border Leaf	Shape egress to ISP commitments; police and re-mark untrusted inbound bursts
AI/HPC Only	Apply ECN; optionally enable PFC where RDMA NICs require lossless treatment

11.8 Border and Management Considerations

Border Leaves sanitize untrusted DSCP, shape ISP egress, and police inbound bursts; OOB management relies on physical/logical separation rather than production QoS.

12 Authentication

Authentication separates identity, authorization, and audit services so current device access and future VPN access share one user base.

12.1 Three-Service Model

1. **OpenLDAP** — Central directory of users, groups, and roles.
2. **TACACS+** — Device access for infrastructure; queries OpenLDAP on each SSH attempt and enforces per-group device access policies.
3. **FreeRADIUS** — VPN and remote user access; co-located with OpenLDAP in the Core Pod.

12.2 User Role Matrix

Eight roles with varying device access permissions and privilege levels:

Group	Device Scope	Priv. Level
super-admins	All fabric and servers	15
net-admins	Spines, Leaves 1–10	10 (network only)
sec-admins	Firewalls, IDS, Bastion	10 (security only)
sys-admins	Admin servers	10 (system only)
hpc-users	HPC servers	5 (limited)
students	Student VMs	3 (minimal)
monitoring	Observability systems	1 (read-only)
external	VPN gateway	VPN access only

12.3 Authentication Flow

SSH access is sent to TACACS+, which checks OpenLDAP identity and group membership, applies device/privilege policy, logs the result centrally, and falls back to local auth only during management-plane failure.

12.4 Security Properties

- Group-based authorization limits each role to approved devices and privileges.
- OpenLDAP changes revoke access on the next authentication attempt.
- All attempts are centrally logged with synchronized timestamps.
- Secrets are stored in OpenBao; steady state should move LDAP/TACACS transport to stronger TLS-protected options where supported.

12.5 Placement

OpenLDAP runs in the Core Pod with Staff and scoped fabric-management reachability; TACACS+ serves the management plane and falls back locally during partitions; FreeRADIUS is co-located for future VPN access using the same directory.

12.6 Addresses Threat T7

Together, centralized identity, group authorization, audit logging, and independent VRF isolation mitigate unauthorized administrative access (T7).